Osmosis and tonicity

Predict water movement from the effective nonpenetrating-solute difference.

Hypotonic

0.10 M outside



Net water enters

Isotonic

0.30 M outside



No sustained net flow

Hypertonic

0.60 M outside



Net water leaves

Outside relative to cell	Animal-cell tendency	Plant-cell tendency
Hypotonic	Swelling; lysis is possible	Water entry builds turgor; the wall resists expansion.
Isotonic	No sustained volume change	Usually less turgid than in a hypotonic environment.
Hypertonic	Water loss and shrinking	Water loss; plasmolysis can occur.

Tonicity depends on nonpenetrating solutes

Assume the same impermeant solute, water-permeable membrane, large external reservoirs and constant temperature. Water still crosses in both directions at balance. Permeating solutes require a time-dependent analysis.

Ideal volume change: $C_{initial} V_{initial} = C_{final} V_{final}$

0.30 M inside -> 0.20 M outside gives $V_{final}/V_{initial} = 1.5$, only if solute amount stays fixed and lysis, pressure differences and membrane tension are ignored.

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Self-check questions and explanations. Refer to the visual guide on the front.

Which outside solutions are hypotonic, isotonic and hypertonic?

0.10 M outside is hypotonic; 0.30 M is isotonic; 0.60 M is hypertonic relative to 0.30 M inside. The same nonpenetrating solute is assumed, so these concentrations predict the osmotic effect.

How do those conditions change an animal cell volume?

Hypotonic: net water entry and swelling. Isotonic: no sustained net flow or volume change. Hypertonic: net water loss and shrinking. Individual water molecules continue moving in every case.

Does water still cross in the isotonic condition?

Yes. At the stated balanced condition water crosses both ways. Equal average opposing fluxes give no net change, not a motionless membrane.

For 0.30 M inside and 0.20 M outside, what ideal volume ratio is predicted?

Water enters. Vfinal/Vinitial = 0.30/0.20 = 1.5. This ideal result assumes solute conservation, no lysis, negligible pressure difference and a reservoir whose concentration remains fixed.

Why does a plant cell respond differently in a hypotonic solution?

Water enters a plant cell and builds turgor pressure. The wall resists further expansion; pressure can balance the osmotic driving force. A wall-less animal cell can swell and potentially lyse.

What changes if the solute itself can cross the membrane?

A permeating solute can redistribute, changing gradients and later water movement. Need relative permeabilities, concentrations and relevant time scale; initial osmolarity alone may not predict lasting tonicity.